MD. SHAHRIAR HOSSAIN

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SUMMARY

Final-semester Computer Science and Engineering student with project experience in LLM applications, computer vision, and data analysis. Builds AI-powered mobile and web interfaces using Python, Kotlin, PyTorch, and TensorFlow, with experience keeping inference and raw training data local.

EDUCATION

North South University, Dhaka | B.Sc. in Computer Science and Engineering

2021 - Present | Final semester | **Expected graduation: December 2026**

TECHNICAL SKILLS

Programming: Python, Kotlin, C, C++, Java, SQL

AI and Data: PyTorch, TensorFlow/Keras, CNN, Transformer, Custom CNN, data analysis, model evaluation

LLMs and Applications: On-device LLMs, LiteRT-LM, multimodal inference, FastAPI, Flask, Streamlit

Development Tools: Git, GitHub, Cursor, Antigravity IDE, Jupyter Notebook, Google Colab

PROJECTS

Solvix AI - Offline On-Device AI Chatbot

Technologies: Kotlin, Jetpack Compose, LiteRT-LM, large language models (LLMs), multimodal inference

- Built an Android AI assistant for local chat and photo questions using LiteRT-LM; inference runs offline after a compatible model is installed.
- Implemented streaming responses, formatted code blocks, and model settings; released a downloadable v1.3 Android build.
- Kept prompt inference on the device, reducing reliance on cloud inference APIs for conversations.

Source code | Android release

Privacy-Preserving Federated and Reinforcement Learning-Based Intrusion Detection System for IoT Devices

Technologies: Final-year capstone project | PyTorch, Flower/FedAvg, reinforcement learning, XAI, Flask

- Analyzed 454,112 cleaned network flows with 42 features and 5 classes; benchmarked 12 ML models, 8 deep learning models, and 5 reinforcement learning agents.
- Implemented Flower/FedAvg training across three clients, exchanging model updates while keeping raw network data local.
- Built a Flask dashboard comparing model predictions and presenting explainable AI (XAI) insights.

Project and results

Explainable AI-Based Eye Disease Classification Using Retinal Image

Technologies: University research | PyTorch, CNN, Transformer, Custom CNN, FedAvg, FedProx, FastAPI

- Evaluated CNN and transformer architectures on 16,242 indexed retinal images across 10 classes using a 2,437-image test split.
- Built a 122,972-parameter custom CNN, explored three-client FedAvg/FedProx training, and connected inference to Streamlit, FastAPI, and Android interfaces.

Project and results | Source code

Languages: Bangla (native), English (professional)